

Student Handout

Module 3: AI as Professional Assistant

Technical Reading with AI: Engineering Mishap Reports

Mission Brief

Today you will use AI as a **professional engineering reading assistant**, not as an authority. Your job is to read, question, verify, and communicate evidence from two real aerospace mishap reports:

1. **Mars Climate Orbiter Mishap Investigation Board Phase I Report**
2. **SOHO Mission Interruption Joint NASA/ESA Investigation Board Final Report**

These are not just “space stories.” They are engineering lessons about units, software interfaces, operations, communication, verification, procedures, and accountability. The Mars Climate Orbiter was lost after a failure to use metric units in ground software used for trajectory modeling [1]. The SOHO report concluded that the interruption was not caused by onboard anomalies, but by ground errors, operational mistakes, inadequate monitoring, and an erroneous decision that disabled part of autonomous failure detection [2].

This session follows the activity structure of the Module 3 activity plan: human skimming, AI-assisted summary, prompt improvement, evidence checking, hallucination detection, professional judgment, discussion, memo writing, and reflection [3].

Learning Aims

By the end of this tutorial, you should be able to:

- Use general-purpose AI tools to support technical reading in engineering.
- Apply a digital mindset when using AI to investigate real-world engineering problems.
- Critically evaluate the strengths, limits, and risks of AI-generated technical summaries.
- Verify AI outputs against primary engineering evidence.
- Communicate technical findings clearly and responsibly.

Course Learning Outcomes

This activity supports:

- **CLO 1:** Utilize modern AI tools to analyze data, solve problems, and support coding in engineering contexts.
 - **CLO 2:** Apply core AI technologies to solve introductory engineering problems.
 - **CLO 3:** Evaluate the capabilities, limitations, and ethical considerations of AI technologies in engineering applications.
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Ground Rules: AI as Assistant, Not Authority

Before using AI, agree with your partner:

1. AI can help you summarize, compare, structure, and question.
2. AI cannot replace evidence.
3. Every important claim must be checked against the report.
4. If the AI sounds confident but gives no evidence, treat it as unverified.
5. Your final memo must be written by you, using checked evidence.
6. In safety-critical engineering, “probably correct” is not good enough.

The activity plan emphasizes that students should “trust evidence, not confidence,” and that every extracted item should include a source reference and manual check [3].

Your Team Roles

Work in pairs.

Role A: AI Pilot

You write prompts, paste excerpts, and ask follow-up questions.

Role B: Evidence Engineer

You check the AI's answers against the reports and record proof.

Swap roles after Mission 3.

Deliverables by the End of Class

You must submit or be ready to share:

1. One upgraded AI prompt.
 2. An evidence log with at least **8 checked claims**.
 3. A table converting report recommendations into engineering requirements.
 4. A hallucination-check table.
 5. A short professional memo.
 6. Exit ticket reflection.
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Reliability Labels

Use these labels throughout the session:

Label	Meaning
A – Directly supported	The report clearly says this.
B – Reasonable inference	The report supports it, but it is not a direct quote.
C – Unclear	More evidence is needed.
D – Contradicted or invented	The report does not support it, or says the opposite.

Mission 0: Launch

Time: 0–5 minutes

Goal: Set up tools, team roles, and responsible AI mindset.

Task

1. Open the two reports:
 - Mars Climate Orbiter Mishap Investigation Board Phase I Report.
 - SOHO Mission Interruption Joint NASA/ESA Investigation Board Final Report.
2. Open your AI tool.
3. Create a shared evidence log.
4. Choose roles: AI Pilot and Evidence Engineer.
5. Write this statement at the top of your notes:

“AI may assist my reading, but I am responsible for verifying technical claims before using them.”

Mission 1: Skim Like an Engineer

Time: 5–15 minutes

Goal: Compare human skimming with AI summary.

The activity plan asks you to compare your own skim with AI output and to trust evidence rather than confidence [3].

Part A: Human Skim First

Without AI, skim the executive summaries or introductory sections of both reports.

For each report, write:

Report	3 things that seem important	2 questions you still have
Mars Climate Orbiter		
SOHO		

Focus on:

- What failed?

- Was the failure technical, human, procedural, organizational, or a combination?
- What evidence does the report give?
- What recommendations were made?

Part B: Ask AI for a Summary

Use this prompt:

I am reading two aerospace mishap investigation reports: the Mars Climate Orbiter Phase I Report and the SOHO Mission Interruption Joint NASA/ESA Report. Using only the excerpts I provide, summarize each report in five bullet points. For each bullet, include:

1. the claim,
2. the evidence phrase I should verify,
3. whether the claim is about hardware, software, operations, communication, management, or verification.

If evidence is missing, say “not enough evidence.”

Paste a short excerpt from each report, such as the executive summary or findings section.

Part C: Compare

Answer:

1. What did you notice that AI missed?
 2. What did AI notice that you missed?
 3. Did AI overstate anything?
 4. What would you need to verify before trusting the AI summary?
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Mission 2: Prompt Upgrade

Time: 15–27 minutes

Goal: Learn that better prompts produce better engineering results.

The activity plan emphasizes that weak prompts produce weak answers, while structured prompts with evidence produce better engineering results [3].

Weak Prompt

Try this first:

What happened in the Mars Climate Orbiter and SOHO failures?

Record one weakness of the AI response.

Upgrade the Prompt

Now improve the prompt using this structure:

- Role
- Task
- Sources
- Output format
- Evidence requirement
- Uncertainty instruction
- Safety-critical caution

Strong Prompt Template

Act as an engineering technical reading assistant. I will provide excerpts from two investigation reports: Mars Climate Orbiter and SOHO.

Your task is to compare the two cases in a table with the following columns:

1. Report
2. Main failure or interruption
3. Direct/root cause
4. Contributing organizational or process factors
5. Evidence phrase to verify
6. What cannot be concluded from the excerpt
7. Reliability rating: high, medium, or low

Use only the text I provide. Do not invent missing facts. If the excerpt does not support a claim, write “not supported by provided text.”

Your Task

Create your own upgraded prompt.

Your upgraded prompt must ask the AI to compare **both** reports and must require evidence.

Write your final upgraded prompt here:

Mission 3: Requirement Treasure Hunt

Time: 27–47 minutes

Goal: Extract recommendations and convert them into engineering-style requirements.

In the Mars Climate Orbiter report, recommendations included verifying consistent use of units and auditing software interface compliance [1]. In the SOHO report, recommendations included reviewing operational procedure changes, validating timelines, verifying spacecraft configuration before critical activity, and reviewing ground system problem reports [2].

Task

Find at least **six recommendations**:

- Three from the Mars Climate Orbiter report.
- Three from the SOHO report.

Convert each into a clear engineering requirement using “shall” language.

Example

Original recommendation:

Verify consistent use of units throughout spacecraft design and operation.

Engineering requirement:

The project team shall verify and document consistent units across spacecraft design, ground software, navigation data products, and operations interfaces before mission-critical events.

Requirement Table

No.	Report	Original recommendation or finding	Your “shall” requirement	Evidence location or phrase	Manual check completed?
1	MCO				Y/N
2	MCO				Y/N
3	MCO				Y/N
4	SOHO				Y/N
5	SOHO				Y/N
6	SOHO				Y/N

AI Support Prompt

Use AI to help organize, not to replace your reading:

I will paste recommendations from two mishap reports. Convert each recommendation into a clear engineering requirement using “shall” language. For each one, identify the likely owner, the verification method, and the risk if ignored. Do not add recommendations that are not present in the text.

Mission 4: Hallucination Hunt

Time: 47–62 minutes

Goal: Detect unsupported or invented AI claims.

AI tools can produce fluent but unsupported claims. Your job is to challenge the answer, label reliability, and catch hallucinations early, as emphasized in the activity plan [3].

Task

Below are “AI-style claims.” Some are accurate, some are incomplete, and some are false. They are intentionally uncited. Verify each one against the two reports.

Label each claim:

- **A – Directly supported**
- **B – Reasonable inference**
- **C – Unclear**
- **D – Contradicted or invented**

Claim	Label	Evidence or correction
1. The Mars Climate Orbiter failure involved a units mismatch in ground software.		
2. The SOHO interruption was caused by a major onboard hardware anomaly.		
3. The Mars Climate Orbiter report identified communication problems between project elements.		
4. The SOHO report found that the operations team failed to adequately monitor spacecraft status.		
5. The Mars Climate Orbiter survived its low-altitude Mars pass and continued its mission.		
6. Both reports show that procedures, verification, and communication can be mission-critical.		
7. The SOHO report recommended ignoring procedure changes if science return is urgent.		
8. The Mars Climate Orbiter report recommended stronger verification and validation of ground software.		

AI Audit Prompt

After AI gives you an answer, ask:

Audit your previous answer. List every claim that needs verification. For each claim, explain what evidence would be needed and whether the claim might be unsupported, overstated, or contradicted.

Professional Reality Check

Time: 47–62 minutes, alongside Mission 4

Goal: Connect reading to safety-critical engineering.

Discuss with your partner:

1. Which case would be easier for AI to summarize? Why?
2. Which case would be riskier to trust AI on? Why?
3. What kind of AI mistake could be dangerous in an engineering workplace?
4. What human checks would reduce that risk?

Use this sentence starter:

In a safety-critical engineering context, I would not act on an AI-generated summary until...

Mission 5: Engineer's Reality Check

Time: 62–70 minutes

Goal: Decide when AI can assist and when engineers must take control.

The activity plan states that students should use AI as support, not authority, because engineers remain accountable [3].

Scenario

Imagine you are on a mission operations team. An AI assistant gives the following advice.

For each item, decide:

- Use
- Use only after verification
- Reject
- Escalate to senior engineer or review board

AI advice	Decision	Why?
“The units are probably consistent, so no need to check the interface file.”		
“The procedure was used before, so changes do not need independent review.”		
“The team should compare navigation results from different methods before a critical maneuver.”		
“If a spacecraft enters safe mode, recover quickly so science operations can resume.”		
“If telemetry disagrees, assume the sensor with the unusual reading is faulty.”		
“Before critical operations, verify configuration against the planned state.”		

Reflection

Complete this sentence:

As an engineer using AI, I am accountable for...

Mission 6: Ed Discussion Exchange

Time: 70–80 minutes

Goal: Share prompts, verification strategies, and lessons learned.

Post to Ed or your class discussion space:

1. Your best upgraded prompt.
2. One AI claim you rejected or corrected.
3. One verification strategy you will use in future engineering work.

Post Template

Best prompt:

One AI claim we challenged:

Evidence we used:

Verification strategy for future work:

Reply to one other team:

- Suggest one improvement to their prompt.
 - Ask one evidence-checking question.
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Mission 7: Reading Memo Sprint

Time: 80–87 minutes

Goal: Turn verified findings into professional technical writing.

Write a short memo of **180–250 words**.

Memo Prompt

Compare the Mars Climate Orbiter and SOHO mishaps as engineering case studies. What common lesson do they teach about using AI as a professional assistant for technical reading? Use evidence from both reports. Explain one advantage of AI and one limitation or risk.

Memo Structure

To:

Tutor / Engineering Team Lead

From:

Your name

Subject:

AI-assisted technical reading lessons from MCO and SOHO

Memo:

Write your memo here:

Required Content

Your memo must include:

- One checked fact from the Mars Climate Orbiter report.
 - One checked fact from the SOHO report.
 - One comparison between the cases.
 - One statement about AI usefulness.
 - One warning about AI limitations.
 - One professional responsibility statement.
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Exit Ticket

Time: 87–90 minutes

Goal: Capture your key takeaway.

Answer individually.

1. One thing AI did well today:

2. One thing AI did poorly or riskily today:

3. One rule I will follow when using AI for engineering reading:

4. Which CLO did today's activity support most for you? Circle one:

CLO 1 / CLO 2 / CLO 3

5. Why?

Evidence Log Template

Use this throughout the session.

No.	Report	Claim	Evidence phrase or section	AI out- put accu- rate?	Reliability label	Notes
1	MCO			Y/N	A/B/C/D	
2	MCO			Y/N	A/B/C/D	
3	MCO			Y/N	A/B/C/D	
4	MCO			Y/N	A/B/C/D	
5	SOHO			Y/N	A/B/C/D	
6	SOHO			Y/N	A/B/C/D	
7	SOHO			Y/N	A/B/C/D	
8	SOHO			Y/N	A/B/C/D	

Final Checklist

Before leaving, make sure you have:

- Used both reports.
 - Written one strong AI prompt.
 - Checked at least 8 claims manually.
 - Converted at least 6 recommendations into requirements.
 - Identified at least one AI hallucination or unsupported claim.
 - Written your memo in your own words.
 - Completed the exit ticket.
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Key Takeaway

AI can make technical reading faster, broader, and more organized. But in engineering, especially safety-critical engineering, speed is not enough. The professional skill is not simply asking AI for answers. It is knowing how to **prompt, verify, challenge, and take responsibility**.